

TASK 2

Story Genre Classification

Experimental journey, training decisions, results, failure analysis, and the path to a production-ready classifier.

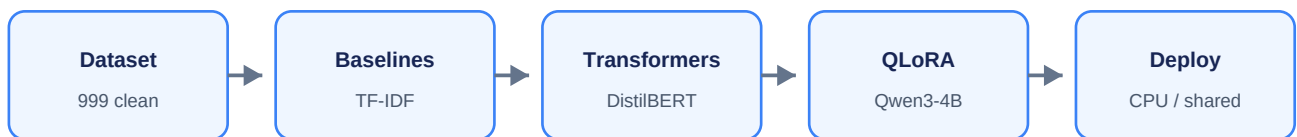
Dataset: 1K Stories / advertised 100 genres

Final clean corpus: 999 stories / 99 observed genre classes

Selected model: Qwen3-4B + QLoRA, checkpoint-100

Final validation: 65.00% Accuracy / 61.97% Macro F1

Experiment -> Measure -> Diagnose -> Iterate



Prepared as a technical assessment report - October 2026

1. Problem Framing

The goal of Task 2 is to classify each story into one of the dataset's genre labels. The assessment encourages reuse of the Task 1 language model where practical, but allows a separate classifier. If the same LLM is adapted, the system must demonstrate that its question-answering behavior is not materially damaged by training.

The problem is unusually difficult for a small dataset: after conservative cleaning there are 999 stories across 99 classes. Most genres have only 10 examples; Fantasy has 9 after removing one malformed record, while Historical Adventure has 20. This leaves roughly ten examples per class before splitting.

Key constraints

Constraint	Implication
99 observed classes	Fine-grained decision boundary; semantically adjacent labels are common.
~10 stories per class	Very limited supervised signal and noisy per-class validation estimates.
Long stories	Single-window transformer truncation can discard discriminative evidence.
Task 1 + Task 2 <= 24 GB VRAM	Full fine-tuning of a large LLM is unattractive; parameter-efficient adaptation is preferred.
Task 3 CPU deployment	Training quality cannot be considered independently of quantization and runtime cost.

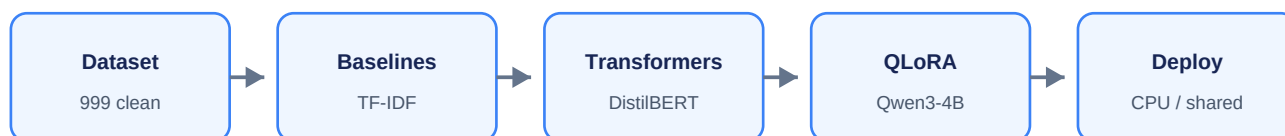
Data quality decision

A conservative scan found one genuinely malformed record: ID 308506, which contained a generation prompt rather than a story. It was removed from the processed corpus; the raw dataset was preserved. Other suspicious short or pattern-matching records were retained after review rather than aggressively filtered.

2. Experimental Journey

The modeling strategy deliberately progressed from cheap, interpretable baselines to more expensive neural approaches. Each experiment answered a specific question rather than simply increasing model size.

Experiment -> Measure -> Diagnose -> Iterate



Stage A - TF-IDF + LinearSVC

A classical text baseline established how much genre signal is available from lexical features. Five-fold stratified cross-validation produced 60.26% mean accuracy and 56.31% mean Macro F1, with approximately 0.162 ms inference per story. On the later fixed train/validation split used for neural comparisons, it achieved 57.50% Accuracy and 52.02% Macro F1.

Decision: keep it as the CPU-speed reference. It is exceptionally fast and surprisingly competitive, but its representation is primarily lexical and does not exploit long-range semantics.

Stage B - DistilBERT with 512-token truncation

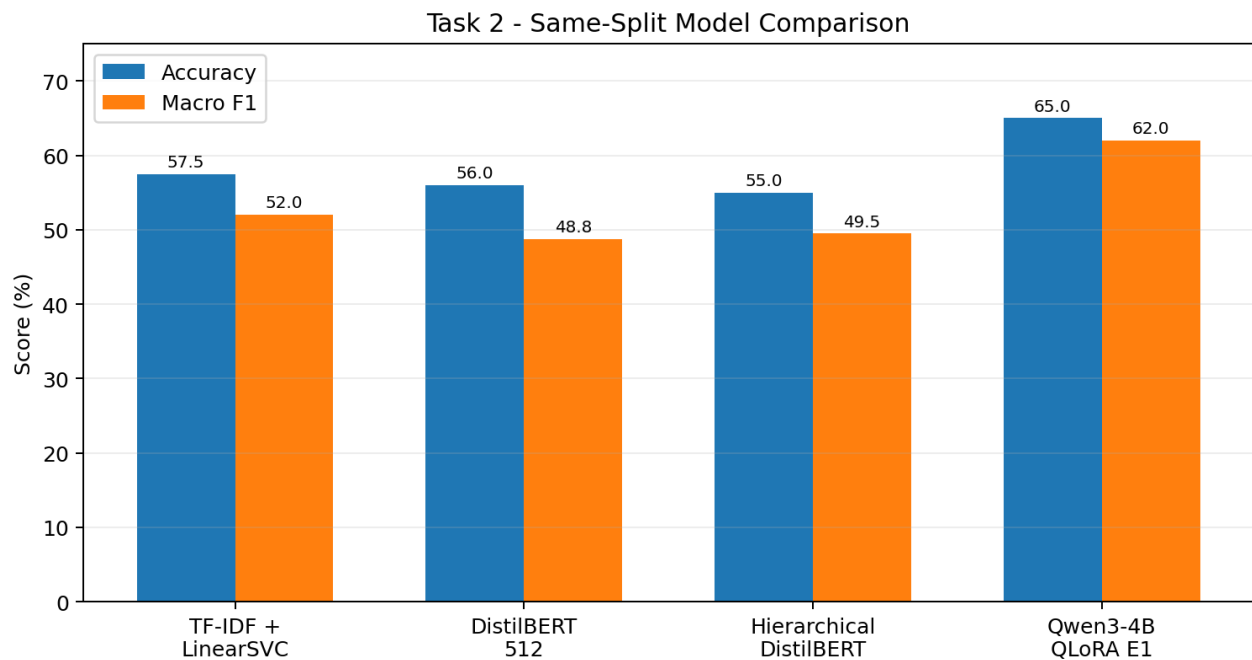
A full fine-tuning experiment tested whether contextual representations would beat the sparse baseline. Best validation results were 56.00% Accuracy and 48.78% Macro F1. The result did not justify the added complexity.

Stage C - Hierarchical DistilBERT

To test whether truncation was the main problem, stories were split into 510-token chunks, each chunk was encoded, and chunk CLS representations were aggregated with masked mean pooling. Best results were 55.00% Accuracy and 49.49% Macro F1. Macro F1 improved only 0.71 percentage points over truncated DistilBERT, while accuracy fell by one point and training cost increased by roughly 3.3x.

Decision: long-document coverage alone was not enough. The dominant difficulty appeared to be the 99-way, low-data classification problem rather than only the 512-token limit.

3. Model Comparison



Model	Accuracy	Macro F1	Role
TF-IDF + LinearSVC	57.50%	52.02%	Fast CPU baseline
DistilBERT, 512 tokens	56.00%	48.78%	Contextual truncated baseline
Hierarchical DistilBERT	55.00%	49.49%	Long-document experiment
Qwen3-4B + QLoRA E1	65.00%	61.97%	Selected quality model

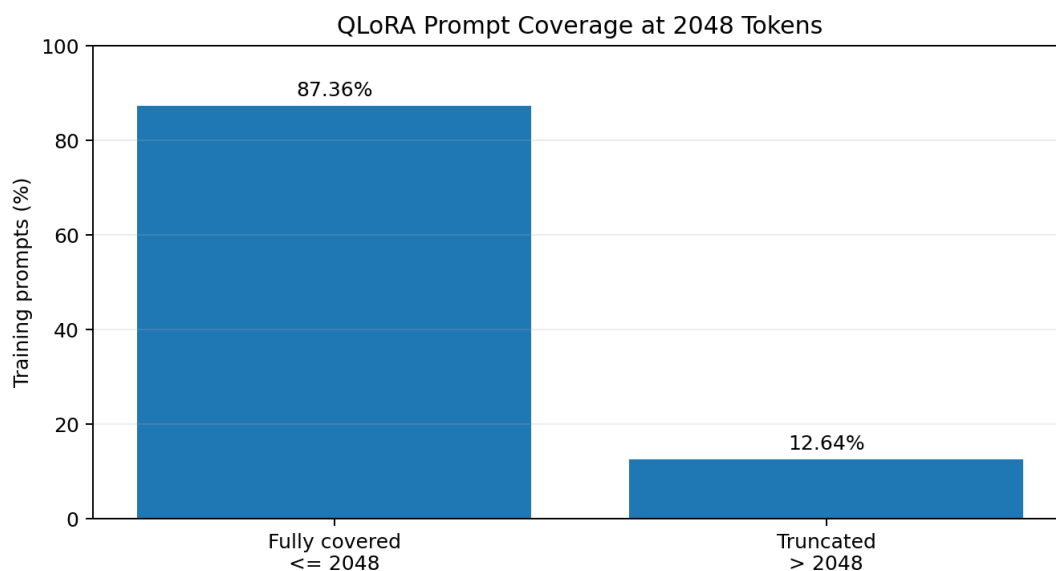
The QLoRA model improved Accuracy by 7.5 percentage points and Macro F1 by 9.95 percentage points over the same-split TF-IDF baseline. This was the strongest evaluated quality result and justified carrying the shared Qwen architecture into deployment experiments.

4. Qwen3-4B QLoRA Experiment

Why QLoRA?

Task 1 already uses Qwen3-4B. Adapting the same base model provides an architectural path toward sharing one quantized base across question answering and classification. QLoRA also keeps the trainable parameter count small enough for a T4-class GPU.

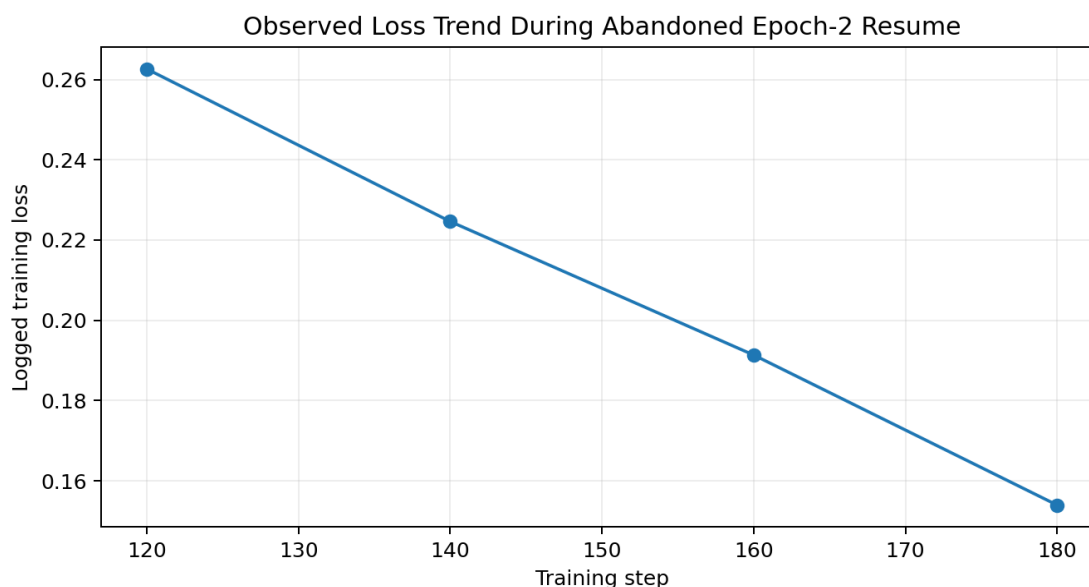
Training component	Configuration
Base model	Qwen/Qwen3-4B
Base loading	4-bit NF4, double quantization, FP16 compute
LoRA	r=16, alpha=32, dropout=0.05
Target modules	q/k/v/o projections + gate/up/down projections
Trainable parameters	33,030,144 / 4,055,498,240 = 0.8145%
Epochs planned	3
Selected checkpoint	checkpoint-100, end of Epoch 1
Batching	batch 1, gradient accumulation 8
Optimizer schedule	LR 1e-4, cosine, 15 warmup steps, weight decay 0.01
Maximum sequence length	2048 tokens
Loss masking	Assistant answer only



87.36% of training prompts fit fully within 2048 tokens. The implementation reserves room for the fixed prompt and target genre, then truncates only the story. This fixed an early bug where naive whole-sequence right truncation produced 100/799 examples with zero supervised target tokens; after the fix, that count became zero.

5. Training Reality and Checkpoint Decision

The first epoch completed successfully and checkpoint-100 was persisted. A later resume toward Epoch 2 reached approximately step 196, but the Colab GPU limit interrupted the run before an additional checkpoint could be saved. Therefore the later in-memory weights were not treated as a valid experiment result.



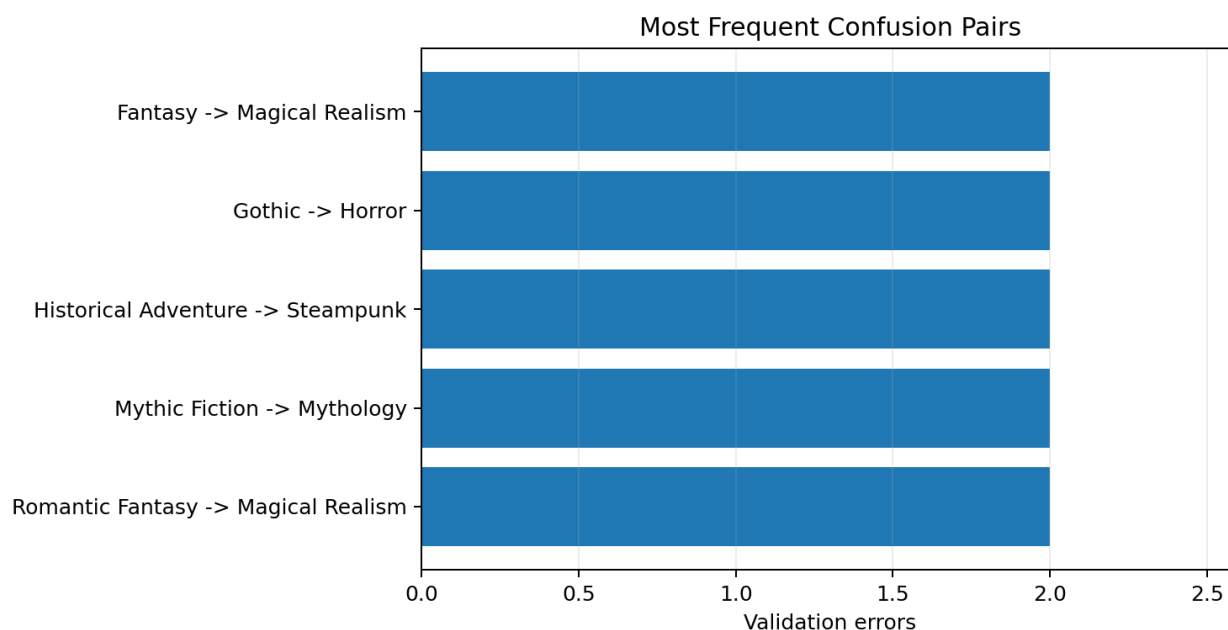
The logged training loss continued to decline during the abandoned Epoch-2 resume (steps 120-180), but this is not evidence that validation quality improved. The selected and reproducible result remains checkpoint-100. Further epochs would require validation-based checkpoint selection.

Reproducible evaluation

Checkpoint-100 was moved to Kaggle and evaluated from a reconstructed, deterministic 799/200 stratified split using seed 42. The final Kaggle run is treated as authoritative because it produced the complete prediction, metrics, classification-report, confusion-matrix, top-confusion, and invalid-output artifacts.

Metric	Final E1 result
Accuracy	65.00%
Macro Precision	65.98%
Macro Recall	65.40%
Macro F1	61.97%
Weighted F1	61.75%
Invalid outputs	5 / 200 (2.5%)
Average T4 inference latency	2.727 s / story
Full validation runtime	9.095 min

6. Error Analysis



Several of the most frequent mistakes are between semantically neighboring genres: Fantasy vs Magical Realism, Gothic vs Horror, Mythic Fiction vs Mythology, and Historical Adventure vs Steampunk. With only about two validation examples per class for many labels, class-level conclusions should be interpreted cautiously.

Invalid generations

True label	Generated output
Supernatural Drama	Mystic Fiction
Humor	Culinary Comedy
Slice of Life	Community Drama
Workplace Drama	Corporate Drama
Experimental	Colorful Fantasy

These five outputs were counted as errors; no post-hoc label mapping was used. For production, constrained decoding or an output grammar over the 99 valid labels is a direct mitigation to test in Task 3.

7. Catastrophic-Forgetting Regression Check

Because the same Qwen family is used for Task 1, classification quality alone is insufficient. A targeted QA regression test compared the quantized base model with the genre LoRA disabled against the same base with the genre LoRA active.

Control	Design
Questions	20 Task-1 QA queries: 15 content QA + 5 long-story QA
Context	Same full-story context for base and LoRA
Generation	Same tokenizer, prompt and deterministic decoding
Coverage	Prompt lengths up to 3,772 tokens; 0 truncated
Manual semantic review	Expected answer preserved on 20/20 tested questions

Result

No factual QA regression was observed on this 20-question subset. The adapted model often produced more concise answers, and in at least one reviewed case it included expected details omitted by the base response.

Scope of claim

This is a targeted regression check, not proof that catastrophic forgetting is impossible. The correct conclusion is: **no regression was observed on the tested 20-question QA subset.**

8. Decisions and Trade-offs

Decision	Why
Preserve TF-IDF baseline	Provides a strong, interpretable and ultra-fast CPU reference.
Do not select DistilBERT	Neither truncated nor hierarchical variants beat the classical baseline on the fixed split.
Stop hierarchical expansion	Only +0.71 pp Macro F1 vs truncated DistilBERT, with ~3.3x training cost.
Select Qwen QLoRA E1	Best evaluated Accuracy and Macro F1; aligns with Task-1 shared-base architecture.
Use checkpoint-100	Only completed, persisted and reproducibly evaluated QLoRA checkpoint.
Count invalid labels as errors	Avoids optimistic post-hoc correction and keeps evaluation honest.
Run QA regression	Required because adapting the Task-1 base model creates a forgetting risk.
Defer deployment optimization	CPU conversion, adapter switching, constrained output and latency/quality trade-offs belong to Task 3.

Selected architecture

The quality path is a shared Qwen3-4B base with a small genre-classification LoRA adapter. The deployment hypothesis is to load one quantized base once and enable the adapter only for classification, while retaining base behavior for Task 1. This architecture still requires empirical validation after llama.cpp/GGUF conversion; the Kaggle NF4 result must not be assumed to survive CPU conversion unchanged.

9. Final Outcome

Task 2 was not a single-model training exercise. It was a sequence of controlled experiments that progressively isolated the real difficulty of the problem: extreme class granularity, low examples per class, long inputs, and the requirement to coexist with the Task-1 QA model.

The final selected checkpoint achieved 65.00% Accuracy and 61.97% Macro F1 on the 200-story fixed validation set, outperforming the same-split TF-IDF baseline by 7.5 and 9.95 percentage points respectively. The model adapts only 0.8145% of Qwen3-4B parameters and showed no observed QA regression on the 20-question targeted check.

What carries into Task 3

Carry-forward item	Task 3 validation
Qwen3-4B base + LoRA	Convert/test shared CPU runtime and adapter switching.
2.5% invalid generations	Test constrained decoding / grammar.
65.00 / 61.97 quality reference	Re-measure after GGUF/CPU conversion.
TF-IDF 0.151 ms/story reference	Keep as latency floor and deployment fallback.
Task-1 regression set	Repeat after deployment conversion.

Task 2 status: complete. The next phase is CPU deployment optimization and end-to-end quality/latency validation.